GigLend Defaulter Handling

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Summary

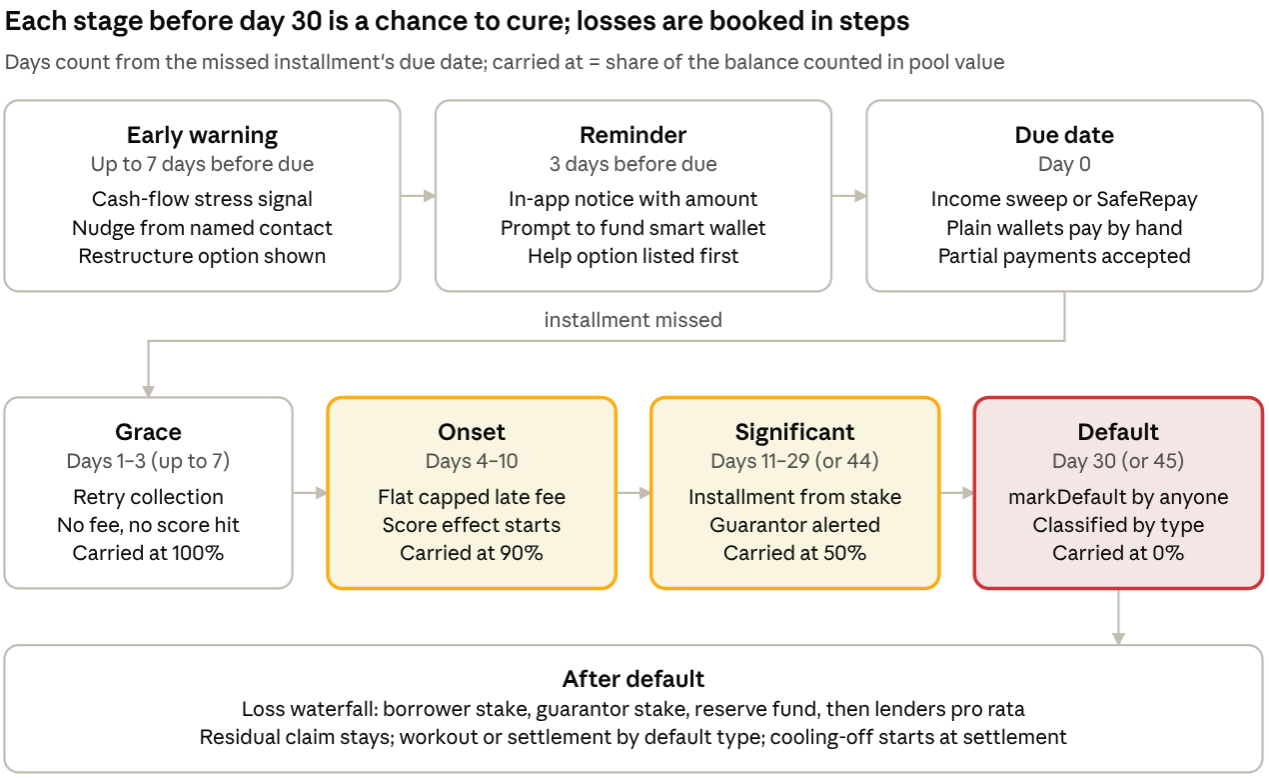
Nothing on a pseudonymous chain forces a borrower to pay, so GigLend spends most of its effort before day 30: lend in ways that make default unlikely, spot stress early, make curing cheap, and recover only from value already locked. When a loan does default, the loss is shared in a fixed order, booked honestly, and handled differently for hardship, strategic default and fraud.

This is the default-handling excerpt of GigLend Project Document, expanded with current research. It is designed for a testnet prototype; Indian rules are a benchmark, not a compliance claim.

1. **Prevent.** Credit-builder loan first, slow capped limit growth, affordability counting all debts, and a pause when first-payment defaults spike.
2. **Detect early.** Watch payout inflows, not just missed payments.
3. **Cure cheaply.** Automatic collection, a no-fee grace window, contact from a named person, and one priced restructure.
4. **Recover from what is locked.** Borrower stake, guarantor stake and income sweep first; then the reserve; lenders last.
5. **Book losses in steps.** Carry late loans at a discount in pool value, so early leavers cannot push losses onto members who stay.
6. **Treat by type, with appeal.** Hardship gets a workout, strategic default a lock-out, fraud no way back.

Three gaps in today's V2 code motivate this: the borrower's stake is not locked, so a default can recover nothing; the 14-day cooldown starts at default instead of at settlement; and a fresh wallet gets a bigger limit than a rehabilitated one.

Default lifecycle



default lifecycle · 7 stages from early warning to default

The first three stages try to stop a miss from happening; the next three try to cure it, with the loan's value in the pool falling as the risk rises. Loans over 3 months, and Trusted or Prime members, get a 7-day grace and a 45-day default horizon. The 90% and 50% carrying values are new in this design and are explained under Loss absorption.

Prevention

A default prevented costs nothing to collect. Seven controls apply before money leaves the pool.

Control	Rule	Why
Credit-builder first	First loan held in escrow at 100% cover; first unsecured loan ₹1,000 at 50% cover, only after 1–2 builder loans and 90 days of stake	At 50% cover a first-loan scammer keeps about ₹1,000 of a ₹2,000 loan; at 100% escrow they keep nothing

Control	Rule	Why
Capped ladder	Next limit at most 1.5× the largest completed loan, smaller early steps, a cap per step, and a fast-climber flag	Evidence from a digital lender: bigger first loans defaulted more and fast climbers behaved strategically (Carlson and Nguyen, via Draft 4)
Total debt burden	Installment at most 35% of assessed income, counting every declared debt; misdeclared debt lowers score and limit	RBI's microfinance benchmark caps all repayments at 50% of household income
Repayment that follows income	Installments match the payout cycle; an income sweep takes 15–30% of each payout	Irregular income, not unwillingness, drives most missed payments [2], [3]
Rule-bound flexibility	Fixed installment; one priced restructure; restructure use tracked in the score	Free flexibility raised strategic default by 50% in a lab-in-the-field study [10]; a priced menu chosen at origination did not raise default in an Indian trial [3]
Fraud screening by cohort	Track first-payment default by week, wallet cluster and band; auto-pause first loans in a cohort that spikes; 5% exposure cap per wallet cluster; weekly growth cap of about 10% of liquidity	A borrower who never makes the first payment signals fraud or a bust-out
Autopay as the default	SafeRepay or the income sweep is set up at disbursement, opt-out rather than opt-in	Commitment devices improve repayment [2]

Autopay without custody is now easier. Since Ethereum's Pectra upgrade went live in May 2025, EIP-7702 lets an ordinary wallet delegate to smart-contract code with spending limits, without moving funds (The Block). A plain MetaMask borrower could therefore opt into the same SafeRepay limits (maximum payment, protected balance, expiry) that the planned GigLend smart wallet will enforce (V2 has none yet). In a real Indian launch the equivalent is a bank mandate such as UPI AutoPay or e-NACH.

Early warning and contact

Act on cash-flow stress before the due date, not after the miss. Indian lenders are moving the same way: an Accion Advisory piece from June 2026 argues for Account Aggregator bank data that triggers action "before repayment stress" rather than "after a missed repayment" (Accion). In the prototype the mock payout stream and wallet balances stand in for that data.

Signal	Trigger (starting value, tune in simulation)	Response
Payout inflows falling	Last 30 days below 70% of the 90-day average	Nudge from the named contact; restructure option shown
Sweep shortfall	Sweep below the installment share for two payouts in a row	Same, plus an offer to switch periodicity
Wallet running dry	Smart-wallet balance below the next installment 3 days before due	Top-up reminder
Fast climber	Two limit steps in 60 days, then a larger request	Hold at the current step
Cohort first-payment default	First-payment default rate in a cohort above its threshold	Pause first loans for that cohort

How to contact

- **A named person, the same one each time.** In a Philippine trial with about 1,259 new borrowers, generic SMS reminders had no significant effect. Only messages naming the loan officer worked: they cut loans unpaid 30 days after maturity by 5.5 points (41%), and only for repeat borrowers that officer had served before (J-PAL). So each member keeps one community contact across loans, and every message carries that name.
- **Let self-curers self-cure.** Many late borrowers pay within days without contact. McKinsey reports that self-cure models freed 5–10% of collector capacity and that better cure-versus-offer decisions cut charge-offs by 10–20% (McKinsey). So after a miss, GigLend's grace window charges nothing and only retries collection; personal contact resumes at Onset.
- **Help first, facts second.** Reminders can raise anxiety under severe stress [2], so each message leads with the restructure option and states the amount and date plainly.

- **Hours and limits.** Contact only 08:00–19:00; no threats, no public posts, no contact with relatives or employers; every notice names the grievance contact. These mirror RBI's recovery directions in force from 1 January 2027 (see Legal guardrails).

Restructuring and hardship

One restructure per loan, asked for before the due date or during Onset, priced, and recorded in the score. A borrower who asks early gets a lighter score effect: the installment counts as late within grace (60%) instead of late (20%).

Option	When	Terms
Payment holiday	Before due, or in Onset	Skip one installment; interest frozen for that period; tenure +1 period; small flat fee
Tenure stretch	Attested income has dropped for two or more payouts	Re-amortise the remaining balance up to the window maximum; the term premium applies
Hardship workout	After a default classified as Hardship	Small fixed installments, interest frozen, at most 6 months; completing it earns back part of the score
Systemic-shock mode	One event hits many borrowers (platform outage, city-wide disruption), confirmed by the multisig and an attested event	Affected cohort only, time-limited: longer grace, no late fees or score effects; principal is never waived

Guardrails:

- At most one restructure per loan and two per member in 12 months, so restructuring is not a routine escape.
- A published no-waiver rule: Indian field research found that borrowers who expect waivers withhold payment (Draft 4, Section 12.5).
- Tailored relief over blanket moratoria: Kenyan lenders that tailored recovery or refinancing did better than those that suspended everyone, though this is an association, not proof (Draft 4).
- Arrears are reported twice, with and without restructured loans, so restructuring cannot hide portfolio stress.

Recovery

Recovery power comes only from value already locked in, or flowing through, GigLend: the borrower's stake, a guarantor's stake, the income sweep and the cost of rebuilding a score. Everything else is persuasion.

Default types

The same default needs different handling depending on why it happened. A simple rule classifies it at markDefault(); the borrower can appeal within 21 days.

Type	Typical signals	Treatment	Settlement discount (max)	Cooling-off after settlement
Hardship (cannot pay)	Asked for help early; payout inflows fell; paid some installments; clean history	Workout plan; smaller scar; path back through a credit-builder loan	25%	30 days after the workout or settlement
Strategic (will not pay)	Payouts kept arriving but installments stopped; no contact or restructure request; stopped soon after a limit increase	Full scar; borrowing locked; stake and guarantor stake applied in full	5%	12 months
Fraud or Sybil (never meant to pay)	First-payment default; new wallet with minimal stake; linked to a flagged wallet cluster	Stake applied; cluster flagged; no rehabilitation	None	Never

The 12-month figure mirrors RBI's 2023 framework for compromise settlements, which bars new credit for 12 months after a settlement (The Wire). The other lengths and the discounts are proposals for the simulation to test.

After default

1. **Residual claim.** The unpaid balance stays on the member record. Half of any future deposit, swept payout or incoming stake is applied to it.
2. **Recoveries refill the reserve first**, then return to the pool, as V2 already does in settleWrittenOffDebt().
3. **Settlement offer.** A lump sum at the discount above closes the claim; recovering most of a balance beats chasing all of it and getting nothing.
4. **Cooling-off starts at settlement**, not at default. Re-entry is always through a credit-builder loan.
5. **Escalating scar.** The scar fades over two years but never below 30% of its size; a second default ends unsecured borrowing.
6. **Permissionless collectors.** Anyone may call collect() or markDefault() and earn a small bounty from the late fee; the contract enforces the timing, so the bounty cannot pull a default forward.

What GigLend will not do

- No public shaming, and no contact with family, friends or employers. Peer visibility is allowed only inside a lending circle whose members agreed to it beforehand, and the evidence for it is a modest gain of about 3.3–3.5% of baseline repayment [11].
- No device locking. RBI allows it only for loans that financed the device, from 30 days past due, with strict limits (Vinod Kothari).
- No sale of claims to outside collectors in the prototype.

Loss absorption

Losses are borne in a known order, recognised as they become likely, and never hidden behind a promise of returns.

Waterfall at default

1. The borrower's locked stake, valued at fair share price (V2 Stage 1; needs the collateral lock, which V2 lacks).
2. The guarantor's locked stake, when the loan has one.
3. The reserve fund (V2 Stage 2).
4. Lenders pro rata, through a lower share price, with the remainder kept as a residual claim (V2 Stage 3).

Book losses in steps (new)

V2 counts a late loan at full value until the moment of default. A lender who withdraws during that window exits at an inflated price and leaves the loss to those who stay. poolValue() should instead carry each loan at a share of its balance that falls with its stage, and restore it if the loan cures:

Stage	Carried at
Current, Grace	100%
Onset (days 4–10)	90%
Significant (day 11 onward)	50%
Defaulted	0%; the loss waterfall takes over

On-chain credit protocols do the same. Goldfinch allows a 45-day grace period, then writes a defaulted loan down linearly over 120 days and reverses the writedown if repayments resume (Goldfinch docs). After its borrower Orthogonal Trading defaulted on 31 million USDC and 3,900 wETH in December 2022, Maple Finance moved to immediate default once a default condition is met and required pool delegates to post first-loss capital (Blockworks).

Other protections

- **Reserve fund.** 10% of repaid interest (V2) as the starting reserve factor, ring-fenced for losses only; target size set by the simulation's stress default rate.
- **Withdrawal queue.** When free cash is short, withdrawals are paid pro rata from a queue, as Maple 2.0 did, so a run cannot drain the pool ahead of recognised losses.
- **Concentration limits.** 2% of liquidity per loan, 5% per wallet cluster and a weekly growth cap, because Maple's 2022 losses came mainly from a single borrower.
- **First-loss layers.** Guarantor stakes now; later, an optional junior tranche of members who take losses first for a higher yield.
- **Disclosure.** The Pool page shows non-performing loans and realised losses every month and never presents returns as guaranteed. RBI's P2P rules ban platform credit enhancement or loss guarantees to lenders, so the reserve must be described as a loss buffer funded by borrower interest, not a guarantee.

Reputation and credit reporting

A default record deters only if the defaulter cannot simply start again as someone new. That makes identity the hinge of the whole policy.

- **On-chain scar.** About 90 points at full strength, fading over two years to a permanent floor of 30% of its size; a second default hits harder and ends unsecured borrowing. Settlement shrinks the scar but never removes it. Every change carries a reason code and can be disputed.
- **One person, one record.** In the prototype the mock attester binds a verified identity (stored only as a hash) to one member record, so a new wallet linked to the same identity inherits the old record. For production, a zero-knowledge Aadhaar proof such as Anon Aadhaar could prove a real, unique person without putting Aadhaar data on-chain. Without this, a fresh wallet beats rehabilitation, as it does in V2 today.
- **Credit bureau reporting.** This is the strongest real-world deterrent, but only regulated lenders report to credit information companies. RBI's 2025 digital lending directions require every loan, including short-term ones, to be reported (IndiaLaw), so a production GigLend would report through a regulated partner.
- **Strategic default mirrors wilful default.** RBI defines wilful default as failing to pay despite the capacity to do so, and gives the borrower 21 days to answer a show-cause notice before a review committee decides (King Stubb & Kasiva). GigLend borrows the definition and the 21-day appeal window; RBI's ₹25 lakh threshold is far above any GigLend loan, so only the process carries over.

Smart-contract implementation

All of this lives in LoanManager and ReserveFund (see the project document). A loan's stage is computed from timestamps when read, so no keeper is needed to move it; stored state changes only when someone acts.

```
enum Stage { Active, Grace, Onset, Significant, DefaultEligible }
enum DefaultType { None, Hardship, Strategic, Fraud }

function stageOf(uint256 loanId) public view returns (Stage) {
    Loan storage l = loans[loanId];
    if (block.timestamp <= l.nextDueDate) return Stage.Active;
    uint256 daysLate = (block.timestamp - l.nextDueDate) / 1 days;
    if (daysLate <= graceDays(l))   return Stage.Grace;        // 3, or 7 for
Trusted/Prime or loans over 3 months
    if (daysLate <= ONSET_END_DAY)  return Stage.Onset;        // 10
    if (daysLate <  defaultDay(l))  return Stage.Significant;  // 30, or 45
for loans over 3 months
    return Stage.DefaultEligible;
}
```

Functions

Function	Caller	Allowed when	Effect
collect(loanId)	Anyone	Stage is Significant and an installment is unpaid	Applies locked stake to the missed installment; pays the bounty
requestRestructure(loanId, option)	Borrower	Before due or in Onset; once per loan; under the per-member limit	Reschedules; emits RestructureGranted
markDefault(loanId)	Anyone	Stage is DefaultEligible and no restructure is in force	Classifies, runs the waterfall, records the residual claim, pays the bounty
settleClaim() payable	Borrower	Residual claim > 0	Refills reserve first, then pool; on full settlement sets coolingOffUntil
settlementQuote(member) view	Anyone	After default	Discounted payoff by default type
startWorkout(loanId)	Borrower	Default type is Hardship	Creates a capped installment plan with interest frozen
appeal(loanId)	Borrower	Within 21 days of classification	Opens a Disputes ticket; the multisig may reclassify
setShockMode(cohortId, until)	Admin multisig with an attested event	Any time	Extends grace and suspends fees for that cohort; never touches principal

Classification rule

```
function classify(Loan storage l, Member storage m) internal view returns
(DefaultType) {
    uint8 fraudFlags;
    if (m.completedLoans == 0 && l.paidCount == 0) fraudFlags++;     //
first-payment default
    if (clusterFlagged[m.clusterId])               fraudFlags++;     //
linked wallets
    if (block.timestamp - m.stakeSince < 30 days)  fraudFlags++;     // new,
thin stake
    if (fraudFlags >= 2) return DefaultType.Fraud;

    bool incomeKeptFlowing = payouts.inflowSince(l.borrower, l.firstMissedAt)
>= l.installment;
    if (incomeKeptFlowing && !l.askedForHelp) return DefaultType.Strategic;
    return DefaultType.Hardship;
}
```

The rule will misclassify some borrowers, so it is public, logged in LoanDefaulted, and appealable.

Parameters (starting values, bounded setters)

Parameter	Starting value
graceDays	3; 7 for Trusted and Prime or loans over 3 months
ONSET_END_DAY	10
defaultDay	30; 45 for loans over 3 months
lateFee	Flat and capped, once per overdue installment, never compounding
bountyShare	10% of the late fee
provision by stage	Onset 10%, Significant 50%, Defaulted 100%
reserveFactor	10% of repaid interest
residualSweepShare	50% of later deposits, payouts and stake
maxSettlementDiscount	Hardship 25%, Strategic 5%, Fraud 0%
coolingOff	Hardship 30 days, Strategic 12 months, Fraud permanent
appealWindow	21 days

Parameter	Starting value
maxRestructuresPer12Months	2

Events

InstallmentMissed, LateFeeCharged, StakeApplied, RestructureGranted, LoanDefaulted(loanId, defaultType, stakeUsed, guarantorUsed, reserveUsed, lenderLoss), ClaimSettled, WorkoutStarted, CoolingOffSet, Appealed, Reclassified, ShockModeSet.

Tests that must pass

- [ ] Dust-stake borrow-and-vanish leaves the attacker with no profit.
- [ ] markDefault() and collect() revert before their day, so a bounty can never pull a default forward.
- [ ] A second restructure on the same loan reverts; shock mode never reduces principal.
- [ ] poolValue() applies the stage provisions, and a withdrawal during Significant gets the lower share price.
- [ ] Full settlement sets coolingOffUntil, and borrowing reverts until it passes.
- [ ] Each default type is produced by its fixture, and an appeal can reclassify it.
- [ ] A new wallet tied to an attested identity inherits that identity's default record.

Legal guardrails

These rules bind regulated lenders, which the GigLend prototype is not; they are used as a design benchmark. The summaries come from law-firm and news write-ups, not the RBI texts, so check the originals before relying on them.

Rule	What it requires	GigLend design
RBI recovery directions (issued 6 Aug 2026, in force 1 Jan 2027)	Contact only 08:00–19:00; no abusive, threatening, anonymous or excessive calls; no posting of borrower details on social media; no intimidation of family, friends or co-workers; call records kept 6 months; a named recovery grievance officer on every communication	Notifier enforces the hours and logs every message; no third-party contact; grievance contact on every notice

Rule	What it requires	GigLend design
Device locking, same directions (Vinod Kothari)	Only for loans that financed the device; partial limits from 30 days past due, full from 60; incoming calls and SOS never blocked; unlock within 1 hour of payment; ₹250 an hour for wrongful locks	Not used
RBI Digital Lending Directions, 2025 (effective 9 May 2025)	Key fact statement with APR; at least a 1-day penalty-free exit; default loss guarantees capped at 5% of the portfolio and invoked within 120 days of default; all loans reported to credit bureaus; complaints escalate to RBI after 30 days	The quote is the key fact statement; free 1-day exit; the reserve is not a guarantee
NBFC-P2P amendments (16 Aug 2024)	No credit enhancement or loss guarantee to lenders, who bear losses; monthly disclosure of non-performing loans and lender losses; no assured recovery	Pool page discloses losses monthly; returns never guaranteed
Compromise settlement framework (June 2023)	12 months before new credit after a compromise settlement	Strategic defaulters: 12-month cooling-off
Wilful defaulter directions (2024)	Wilful default = can pay but does not; 21 days to answer a show-cause notice	Strategic class uses the same test; 21-day appeal

Sources

Web pages opened on 25 Oct 2026. Numbers in brackets refer to the GigLend literature review: [2] Al Maruf et al. 2024, [3] Barboni and Agarwal 2023, [10] Czura, John and Spantig 2026, [11] Ahmed 2026. "Draft 4" is the GigLend project design PDF.

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